

XIAOYU ZHOU

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EDUCATION

Peking University

Ph.D. in Computer Science and Technology; GPA: 3.80/4.00

Advisor: Yongtao Wang

Beijing, China

Sep. 2023 – present

University of Electronic Science and Technology of China

B.E. in Electronic Information and Software Engineering; Rank: 1/113

Advisor: Xing Xu

Chengdu, China

Sep. 2019 – Jun. 2023

RESEARCH PROFILE

My research focuses on 3D vision, spatial intelligence, and world model, with an emphasis on learning expressive 3D/4D representations for scene reconstruction, controllable generation, simulation, and open-ended understanding. I am particularly interested in geometry-grounded generative and reconstructive models that bridge visual synthesis and the physical world, aiming to build realistic, interactive, and self-evolving 4D intelligent systems.

PUBLICATIONS

- **Xiaoyu Zhou**, Zhiwei Lin, Xiaojun Shan, Yongtao Wang, Deqing Sun, Ming-Hsuan Yang. DrivingGaussian: Composite Gaussian Splatting for Surrounding Dynamic Autonomous Driving Scenes. *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. | [Link](#)
- Yajiao Xiong*, **Xiaoyu Zhou***, Yongtao Wang, Deqing Sun, Ming-Hsuan Yang. DrivingGaussian++: Towards Realistic Reconstruction and Editable Simulation for Surrounding Dynamic Driving Scenes. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2025. | [Link](#)
- **Xiaoyu Zhou**, Xingjian Ran, Yajiao Xiong, Jinlin He, Zhiwei Lin, Yongtao Wang, Deqing Sun, Ming-Hsuan Yang. Gala3D: Towards Text-to-3D Complex Scene Generation via Layout-guided Generative Gaussian Splatting. *International Conference on Machine Learning (ICML)*, 2024. | [Link](#)
- **Xiaoyu Zhou**, Jingqi Wang, Yongtao Wang, Yufei Wei, Nan Dong, Ming-Hsuan Yang. AutoOcc: Automatic Open-Ended Semantic Occupancy Annotation via Vision-Language Guided Gaussian Splatting. *International Conference on Computer Vision (ICCV)*, Highlight, 2025. | [Link](#)
- **Xiaoyu Zhou**, Jingqi Wang, Yongtao Wang, Deqing Sun, Ming-Hsuan Yang. EA3D: Online Open-World 3D Object Extraction from Streaming Videos. *Neural Information Processing Systems (NeurIPS)*, 2025. | [Link](#)
- **Xiaoyu Zhou**, Zhiwei Lin, Xiaojun Shan, Yongtao Wang, Deqing Sun, Ming-Hsuan Yang. SAMPLING: Scene-adaptive Hierarchical Multiplane Images Representation for Novel View Synthesis from a Single Image. *International Conference on Computer Vision (ICCV)*, 2023. | [Link](#)
- **Xiaoyu Zhou**, Xiaotong Song, Hao Wu, Jingran Zhang, Xing Xu. MAVT-FG: Multimodal Audio-Visual Transformer for Weakly-supervised Fine-Grained Recognition. *ACM Multimedia (MM)*, 2023. | [Link](#)

- Wenyuan Yang, **Xiaoyu Zhou**, Zhikai Chen, Bofei Guo, Zhongjie Ba, Zhihua Xia, Xiaochun Cao, Kui Ren. AVoid-DF: Audio-Visual Joint Learning for Detecting Deepfake. *IEEE Transactions on Information Forensics and Security (TIFS)*, 2023. | [Link](#)
- Jingtong Yue*, Zhiwei Lin*, Xin Lin, **Xiaoyu Zhou**, Xiangtai Li, Lu Qi, Yongtao Wang, Ming-Hsuan Yang. RobuRCDet: Enhancing Robustness of Radar-Camera Fusion in Bird's Eye View for 3D Object Detection. *International Conference on Learning Representations (ICLR)*, 2025. | [Link](#)
- Xin Lin*, Shi Luo*, Xiaojun Shan, **Xiaoyu Zhou**, Chao Ren, Lu Qi, Ming-Hsuan Yang, Nuno Vasconcelos. HQGS: High-Quality Novel View Synthesis with Gaussian Splatting in Degraded Scenes. *International Conference on Learning Representations (ICLR)*, 2025. | [Link](#)

Note: * denotes equal contribution.

RESEARCH EXPERIENCE

Wangxuan Institute of Computer Technology, Peking University Beijing, China
Research Assistant Feb. 2023 – Sep. 2023

- Conducted research on the reconstruction, generation, understanding, and simulation of 3D/4D scenes. Proposed a series of works to build world simulation systems for physical AI and spatial intelligence.

National Key Laboratory of Big Data Chengdu, China
Research Assistant Mar. 2022 – Oct. 2022

- Conducted research on foundation vision encoding models and multimodal representation learning using vision transformers, and applied the learned representations to diverse downstream vision tasks, including detection, segmentation, and fine-grained recognition.

Communication and Information Security Labs Shenzhen, China
Research Assistant Apr. 2021 – Nov. 2022

- Conducted research on trustworthy AI, focusing on adversarial defense and automated detection of deepfake technologies.

WORK EXPERIENCE

ByteDance Beijing, China
Research Scientist Intern Feb. 2026 – Jun. 2026

- Conducted research on foundation vision models, with an emphasis on long-sequence scene generation; developed a 4D representation-based memory space to improve the spatial consistency of scene synthesis.
- Built an efficient pipeline for data processing and feature analysis of large-scale web data, including data cleaning, alignment, and quality analysis.

Changan Automobile Group Co., Ltd. Remote
Research Scientist, Collaboration Project Dec. 2024 – Sep. 2025

- Developed an automated 3D data annotation pipeline for autonomous driving and designed a reconstruction-based framework for generating open-ended semantic occupancy.
- Designed an autonomous driving simulation engine and proposed a 3D Gaussian Splatting-based algorithm for simulating real-world driving scenes.

AWARDS & HONORS

National Scholarship: First Prize, top 1% in 800 students.

Peking University Presidential Scholarship: Top honor for Ph.D. students.

SenseTime Scholarship: 1 out of 30 AI researchers worldwide.

Alibaba Taotian Scholarship: 1 out of 15 AI researchers nationwide.

FunTech Science Innovation Scholarship: 1 out of 20 AI researchers nationwide.

Academic Innovation Award, Peking University: Top 3% in 100 students.

International Mathematical Contest in Modeling: Finalist, top 1% worldwide.

Internet Innovation and Technology Competition: Gold Medal, top 0.1% nationwide.

National Information Security Competition: Second Prize, top 0.5% nationwide.

National Computer Design Competition: Second Prize, top 0.7% nationwide.

International Innovation Competition: Bronze Medal, top 1% nationwide.

PROFESSIONAL SERVICE

Conference Reviewer: CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, SIGGRAPH.

Journal Reviewer: TPAMI, IJCV, TIP, PR.